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Teen And Up Audiences

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The Citicorp Tower

by DeltaFell

Summary

Last year, Adora designed the new Citicorp Tower, with a revolutionary new base style to account for some interesting design challenges. It was truly a feat of modern engineering, if you were to ask her. Last month, an engineering student by the name of DT reached out to her. They were writing a thesis on her design, and discovered it would be quite likely to fail. Adora ran her calculations again, and asked Spinnerella to run wind tunnel simulations of the building. As it turns out, some moron at the construction company had decided to replace the welds along important structural elements with bolts, weakening it.

Notes

See the end of the work for notes

Adora straightened at her desk. The results from the simulation were back, and things did not look good. The quartering winds, combined with the improperly reported change to bolts meant that the Citicorp tower would fail under a 16-year storm. Which New York was due for one, this August. Shit.

DT was right, holy shit they were going to be fucking insufferable for her class.

If only she didn't have to account for that freaking church, it would have been fine. If the contractors hadn't bid too low and stupidly decided to cheap out on the build by replacing welds with bolts, everything would have been just fine. But no, they had to go cheap on her.

Adora sighed, she had better start calling someone. But who to call? If she let news of this mistake slip to the public, she was ruined as an engineer. It would cause a panic and she'd never have work again. Clearly, the press was not an option. She pinched the bridge of her nose. She should call her lawyer. Glimmer would know what to do.

Adora picked up her office phone and dialed Glimmer's office.

"Hello?" said Glimmer.

"Hey Glim, it's Adora. We need to talk about the Citicorp building," said Adora.

"Sure, whatcha need? Those suits stiff you? I know a good place for bodies," Glimmer said.

"If New York gets hit by a hurricane this summer, it will collapse, killing everyone in the surrounding neighborhoods."

"Well shit. How the hell did you screw up so badly Adora?"

"Well I didn't account for quartering winds in the initial calculations, and in my defense, we aren't required to by the state. Just that wouldn't have done it though. The stupid contractor replaced the welds on the chevrons with bolts, which fucked up the stresses massively. What am I going to do?" said Adora.

“I have absolutely no idea what that means, but I assume it's bad? Just remember Adora, we have your back here,” said Glimmer

“If word gets out, I’m ruined." Adora held her face in her palms. "Nobody will ever hire me as an engineer again, plus the building might not even get fixed if we don't do it right."

"Okay, can we maybe talk to Citicorp? Like reach out to their lawyers or something?" Glimmer said.

"Maybe, we just have to hope that we can reach someone who'll understand and actually prioritize safety," Adora said. "Goddess forbid we run into someone wearing their fucking manager hat."

"I don't know what that means, but I think I get the idea. Okay, I'll talk this over with Bow and get back to you tomorrow, I'll call you then," said Glimmer.

"Okay. Thank you, Glimmer, really. I appreciate you taking this so seriously," said Adora.

"Of course! Now bye, I gotta go," Glimmer said and hung up.

Adora hadn't slept well that night, and she was feeling it today. Her anxiety had kept her up almost all night, and when Glimmer didn't call by ten, she started to feel sick to her stomach. Her mouth felt dry, as if she was stuck in the Mojave.

Adora knew she would not get any work done, but she tried anyways. She managed a scant few lines drafting up an emergency fix before she had to set it aside.

By lunchtime at noon, Adora's anxiety had increased tenfold, and her hands shook. She wanted to go home and curl up into a ball,

but she had to stay. What if they called right after she left?

At one, her phone rang. Adora jumped, staring at it incomprehensibly for a few seconds before scrambling to pick it up.

"H-hello?" she said.

"Hey Adora," Catra said.

"Catra? How the hell did you get this number?" Adora asked.

"Oh I got it from that sparkly lawyer of yours," said Catra.

"Glimmer? But - wait, are you working for Citicorp now?" said Adora.

"Yup," Catra said, "I'm the CEO even."

"CEO? What happened to wanting to be an engineer?" Adora said.

"It's a long story, but I am actually a licensed engineer. I heard you built us a flawed building? What gives, Adora?" said Catra.

"Okay, first of all, if the contractors had just followed my plans, then this wouldn't be an issue. But since they swapped out the welds on chevrons for bolts, all the stresses are way higher," said Adora.

"How much higher?" asked Catra.

"Forty percent in simulations, more in the real world," said Adora.

"What was the safety factor used?"

"The firm used the truss standard of 1:1, rather than 1:2."

"Oh fuck," said Catra.

"Yeah, that's what I said."

"Wait, you said fuck? My my, how the times have changed," said Catra.

"Yes, I say fuck now when the occasion calls for it, you'll just have to get use to it," Adora said.

"Oh I wasn't complaining, princess. So, do you have any suggestions for the fix yet?"

"No, I was hoping to hear back from Glimmer before committing to any one choice of action," said Adora.

"Y'know what? Come on up to my office tomorrow, we'll work on it together," said Catra.

"Just like old times?" asked Adora.

"Just like old times," said Catra.

Adora had slept much better that night. Time with Catra had always made her sleep better at night, even if she'd had a shitty day.

Catra's office was in another skyscraper in New York City. It would have felt imposing on any other day, with any other woman waiting for her inside, but it didn't even phase Adora today.

Carrying her calculations, blueprints, and multiple pencils, Adora walked straight for the elevator when she entered the 339 Park Ave office building. Normally Adora would have stopped nervously at the front desk, but it was Catra up there. She wasn't going to stop for something as small as social niceties.

Adora boarded the elevator along with a tall, buff woman with short white hair, and a shorter, stockier woman with blue-green hair and an eyepatch.

The white-haired woman tried to start up a conversation with her. “So, who are you? I’ve never seen you around here before.”

“I’m Adora Gray. I’m here to see Catra,” said Adora.

“Oh, first name basis, wow. She doesn’t let anyone in that close. Are you two close friends? Uh, I’m Scorpia, by the way,” she said.

“Yeah, old friends. It’s been a while,” said Adora.

“So what’re all the papers for?” asked Scorpia.

“I’m afraid I can’t tell you that,” Adora said.

“Oh,” said Scorpia, “okay then.”

The woman with blue-green hair snorted.

The rest of the elevator ride was silent, save for the smooth jazz playing from the speakers. The woman with blue-green hair got off first, and then Scorpia got off on the floor before Catra’s.

“Good luck with whatever it is that you’re doing!” Scorpia said as she stepped out.

The doors closed, and Adora pushed the button for Catra's floor again. She waited only a short time before the doors opened up once more, but it felt like forever.

Adora stepped out of the elevator and looked around her. According to the handy little sign placed conveniently directly across from the elevator, Catra's office should be to the left of the elevator.

Adora turned left and began checking doors, bit before she got too far, she ran into Catra.

"Catra!" she said.

"Hey Adora, just the girl I was looking for. C'mon," said Catra, grabbing her hand, "let's use a conference room for this, more table space for all your papers."

Adora followed along as Catra pulled her through the hallways. After accidentally walking in on a few meetings happening in supposedly unoccupied rooms left them giggling, they eventually found an unused conference room.

A long, vaguely oval shaped table occupied much of the room, surrounded by chairs on all sides with a projector cart off to one side.

"Perfect," said Catra, "go on ahead and lay your papers out, Adora."

Adora stretched out the blueprints and set her calculations off to one side.

"So, where are the issues, Adora?" asked Catra.

"Well, you know about the bolts in the chevrons, obviously," Adora circled the offending joints, "but we also should worry about the tuned mass damper."

"Really? That's new," said Catra.

"So, with the tuned mass damper active, the kind of storm that could topple the building would happen about once every 55

years. But, stronger storms tend to knock out the power, and without the tuned mass damper active, a 16-year storm could collapse it, rather than a 55-year storm," said Adora.

"Okay, so we should definitely address that," said Catra.

"Yes," said Adora, "I think emergency backup generators for the mass damper will solve that problem."

"I'd agree with that, yeah. All things considered, that'll probably be the least expensive fix to make," said Catra.

"Well, whatever we decide to do, we should put strain gauges here, here, and here," Adora said, marking the highest stress members, "so we can monitor the situation as the repairs move on."

"I would agree with that. Do you have any ideas as to how we can address the bolted joints?" asked Catra.

"I was thinking maybe we add trusses at the midpoints of the chevrons, and weld them at the joints. It would be expensive, and not very pretty, but I think it would work," said Adora.

"That requires external work though, which I'd rather avoid if we can." Catra scratched her chin. "What if we just weld plates on top of the bolted joints? That should work, won't it?"

Adora immediately leaned over her calculations and began sketching some examples. As she ran the numbers, Catra leaned over her shoulders a few times to see what she was doing.

As Adora finished, she looked up and turned her head to see Catra's face right next to hers. Out of old habit, Adora kissed Catra right away. As soon as she processed what she was doing however, Adora pulled away immediately.

"Oh my gosh, I'm so sorry, I didn't mean-" Adora apologized rapidly before she saw the hurt look on Catra's face -"Catra? What's wrong?"

"Did you not want to kiss me, Adora?" asked Catra.

"No, I do, I want to so much, but I didn't get your consent first. It's been so long, and we haven't seen each other since freshman year, so y'know," Adora said.

"Oh shut up and kiss me, princess," Catra said, pulling Adora into a kiss.

They held the kiss for several minutes, eventually breaking for air.

"So, got any news for me, Adora?" asked Catra.

"So these are just preliminary calculations, but I think your suggestion would work quite well. It might even hold up to a 700-year storm if we use the right materials," said Adora, giving Catra a peck on the cheek.

"I'll get Entrapta to run more simulations on that tomorrow, so we can find the right plates to weld. Even better though, these plates would be on the inside of the building, so our reputations would remain unblemished," said Catra.

"What about the tenants?" said Adora.

"We'll just do the repairs at night while everyone's at home," said Catra.

"Devious. I like the way you think," said Adora.

"I know you do," Catra said.

Catra and Adora met with city officials and prepared an evacuation plan for up to 200,000 people just in case something went wrong during the modifications

Months later, the Citicorp building had been successfully modified to withstand the stresses of hurricane force winds. About six weeks into the secretive modification process, Hurricane Prime had threatened New York City, however, it turned away at the last minute, and the evacuation plans never needed to be used.

A few months after the resolution of the crisis, Catra and Adora decided to get married. Many of their friends and coworkers were invited, and among the wedding gifts was a scale model of the new Citicorp building from Entrapta accompanied by a note only saying "I told you so," a pair of brand new devices called calculators from Glimmer and Bow, and lastly an intro to structural engineering textbook from DT.

End Notes

I'm sorry I'm late today was exceptionally busy, I spent all day bouncing between working and trying to find new glasses, because my current scrip has been giving me splitting headaches since at least september, probably longer. This was an idea I had back in like June(?) from one of my classes, and I finally made it happen.

"manager hat" is a derogatory term that entered engineering parlance after the challenger explosion, as the phrase "you need to take off your engineer hat and put on your manager hat" was said in the meeting where the shuttle was cleared for that launch.

This fic is based off the actual Citicorp Building incident, which happen in the 1978 with structural engineer William LeMessurier. In my class, I recall being told that part of the reason this ended well was the fact that the Citicorp CEO had some engineering experience and so he was able to get why it was important to deal with this, but I didn't see that in the research I did while writing this, so take that with a pinch of salt. The numbers and specific fix (plus the close call with the hurricane) are actually how things turned out in the real incident, albeit everything is sped up to yeet m/s because lesbians.

Review from an actual engineer: "Why? Why would you do this?"

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